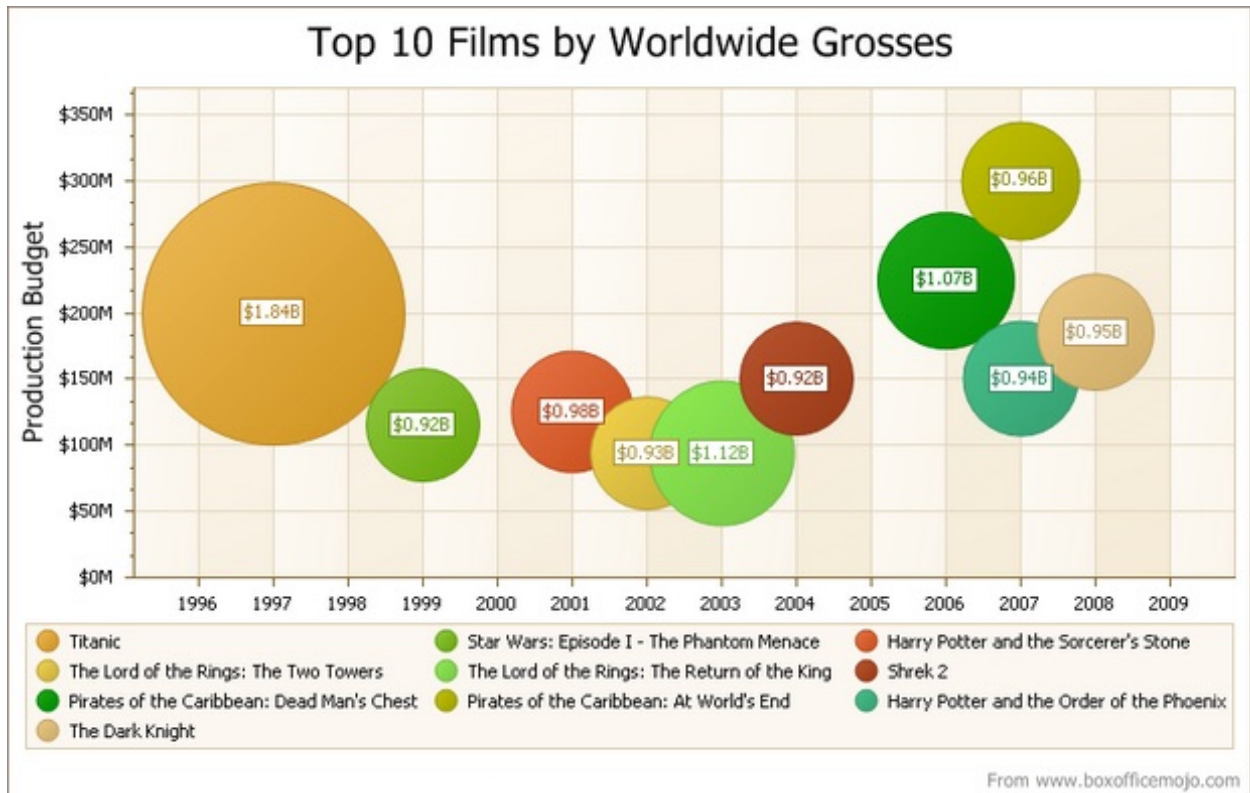


Assignment 9

2025-12-15

Top 10 Films by Worldwide Grosses

```
knitr::include_graphics("bubbles.jpg")
```



Main Problems in the Graph

- 1) Area and Proportion Problem

In this graph, the bubble sizes can be misleading. For example; Titanic earns more than twice The Lord of the Rings: The Return of the King. But, when we look actual numbers, Titanic's revenue is not even two times higher than The Lord of the Rings: The Return of the King. This can't true comments.

- 2) Similar Colors

Another problem is use of similar colors. The colors are so close together that we can't easily read this graph. For example; Shrek 2 and Harry Potter and Sorcerer's Stone. Or Star Wars: Episode 1- The Phantom Menace and The Lord of the Rings: The Return of the King.

Suggestions for Improvement

- 1) Better Proportional Scaling

Values should be shown more true. If one film has twice revenue another film, the bubble size in the graph should also look around two times bigger. This would make visualization more realistic.

- 2) Transparency

Making the bubbles slightly transparent can improve readability, especially when bubbles overlap.

- 3) Color Choices

More distinct colors can be used. Films from the same series can have similar colors but different shades, while different films can be shown with clearly different colors.

Alternative Graph

Instead of this graph, a scatter plot can be used.

X-axis: Release year

Y-axis: Worldwide gross

The budget and abbreviated versions of the film titles are written above the dots. But, in this graph, two films released in 2007 may overlap. This problem can be solved by slightly moving one of the points to the left or right.

I cant give original link for graph. because, this graph is an archieved image from the old Box Office mojo website design. It's not currently live as the site has been updated. However, this graph usually refereneced under the category of 'misleading graph'.

I shared the link where I found the graph and the link to the photo.

<https://cdn0.tnwcdn.com/wp-content/blogs.dir/1/files/2015/05/viz8.jpg>

<https://thenextweb.com/news/7-most-common-data-visualization-mistakes>